

Faecal carriage of extended spectrum beta-lactamase (ESBL) and New Delhi metallo beta-lactamase (NDM) producing *Escherichia coli* between piglets and pig farmworkers

Shikha Tamta^{a,1}, Vinodh Kumar O.R.^{a,1,*}, Pruthvishree B.S.^b, Karthikeyan R.^a, Ramkumar N. Rupner^a, Chethan G.E.^c, Z.B. Dubal^d, D.K. Sinha^a, B.R. Singh^a

^a Division of Epidemiology, ICAR- Indian Veterinary Research Institute, Bareilly, India

^b Veterinary Clinical Complex, Veterinary College, Gadag, Karnataka, India

^c Department of Veterinary Medicine, College of Veterinary Sciences and Animal Husbandry, Central Agricultural University, Selesih, Mizoram, India

^d Division of Veterinary Public Health, ICAR- Indian Veterinary Research Institute, Bareilly, India

ARTICLE INFO

Keywords:

carbapenem resistance
plasmid
E. coli
piglets
farmworkers
India

ABSTRACT

A cross-sectional study on five organized pig farms was conducted to assess the faecal carriage of ESBL and blaNDM carbapenemase-producing *E. coli* in piglets and pig farmworkers. Faecal samples from piglets (n = 155) and pig farmworkers (n = 21) were processed for isolation and characterization of *E. coli*. A total of 124 *E. coli* isolates from piglets and 21 *E. coli* isolates from pig farmworkers were recovered and screened for ESBL production. The results showed that 44.4 % (55/124) of the isolates from piglets and 42.9 % (9/21) of the isolates from farmworkers were ESBL positive. The ESBL positive isolates from piglets and farmworkers harbored blaCTX-M and also co-harbored other beta-lactams, sulphonamide, quinolone and tetracycline resistance genes. Diarrhoeic (50%, 49/98) and crossbred piglets (52.7%, 39/74) harbored a significantly higher number of ESBL producing isolates than non-diarrhoeic (23.1 %, 6/26) and purebred piglets (32%, 16/50) (p < 0.05). Piglets and pig farmworkers harbored nine and two carbapenem-resistant isolates, respectively. Interestingly, two isolates from piglets and one isolate from farmworkers harbored the blaNDM gene. The blaNDM positive *E. coli* isolated from piglets and farmworkers of the same farm revealed similar antibacterial resistance patterns, resistant genes, sequence (ST-167) and plasmid type (IncX3). In India, carbapenems are not used in food animal treatment, hence carbapenem resistant *E. coli* in piglets possibly originated from the human contact or common environment and is of public health importance.

1. Introduction

Antimicrobial resistance (AMR) is a global problem and resistance in *E. coli* is a critical threat to public health. Extended-spectrum β -lactamases (ESBLs) can hydrolyze most of the β -lactams including the fourth generation cephalosporins except cephamycins and carbapenems. Carbapenemases are the most powerful beta-lactamases; hydrolyze almost all available beta-lactam antibiotics [1]. The emergence of carbapenem resistance is critically important for physicians since carbapenems are the last-line antibiotics available for treating life-threatening infections caused by Gram-negative bacteria and hence, their use in food animals is not licensed [2]. Recently, carbapenem resistance is emerging

globally because of the acquisition of carbapenemase genes or by high-level ESBL expression combined with the loss of porins [3]. Emerging reports suggest that third-generation cephalosporin-resistant bacteria associated infections cause decreased treatment efficacy [4] and carbapenems are also less reliable as last-resort antibiotics because of increasing resistance [5]. Reports are available on the presence of carbapenem-resistant and blaNDM gene in *E. coli* isolated from humans, however, limited reports exist on carbapenem-resistant *E. coli* from animals [6–10]. The carbapenem-resistant, ESBL producing Enterobacterales are classified in the priority list -1 pathogens and need immediate intervention to decrease their emergence [11]. The emergence of AMR in bacteria of livestock populations is connected to the

* Corresponding author.

E-mail address: vinodhkumar.rajendran@gmail.com (V.K. O.R.).

¹ Shikha Tamta and Vinodh Kumar O.R. contributed equally and are joint first authors.

emergence of AMR in bacterial populations that colonize and infect humans [12]. The objective of the present study was to determine piglets and farmworkers as a potential source of ESBL and blaNDM carbapenemase producing *E. coli*.

2. Materials and methods

A cross-sectional study was conducted from July 2016 to April 2017 and designed to sample five organized pig farms covering three pig-breeding states of India namely Uttar Pradesh, Karnataka, and Tamil Nadu, and to characterize the carbapenem resistant and ESBL producing *E. coli* isolates in piglets and farmworkers. A semi-structured peer-evaluated questionnaire was used to collect the information about swine farm location, latitude, longitude, altitude, size of the farm, use of antibiotics, number of farmworkers and the prevalence of piglet diarrhoea. EpiTools software (<http://epitools.ausvet.com.au/content.php?page=h>) was used to calculate the minimum number of samples to be collected in each farm based on the prevalence of 10 to 20 % (confidence interval - 95 percent; power - 0.80; margin of error - 5 %) from our earlier studies [8–9,13–14]. A case was considered diarrhoeic when the piglets pass watery faeces more than four times a day, for at least one day. The piglets not treated with antibiotics at least a fortnight preceding the sample collection were sampled. Faecal samples from both diarrhoeic (n = 112) and non-diarrhoeic piglets (n = 43) of <3 months of age along with healthy pig farmworkers' faecal samples (n = 21) were sampled once in the study period (Table 1). The health status/ recent treatment history of the farmworkers was also collected. The faecal swabs (HiMedia, Mumbai, India) were collected from individual piglets and transported to the laboratory under cold chain for isolation and characterization of *E. coli*.

2.1. Isolation and identification of *E. coli*

After pre-enrichment of faecal swabs in buffered peptone water (incubation at 37 °C for 2–4 h) and enrichment in MacConkey broth (incubation at 37 °C for 2–4 h), the selective plating of faecal swab was performed on MacConkey agar supplemented with cefotaxime (1 mg/L) and incubated at 37 °C for 12 h. The isolated colonies were picked up, streaked on eosin methylene blue (EMB) agar medium, and incubated at 37 °C overnight. Subsequently, the colonies displaying a characteristic metallic sheen on EMB were picked for biochemical tests. The reference strains obtained from the Division of Epidemiology, Indian Veterinary Research Institute, Izatnagar repository were used in this study as positive controls (GenBank accession number KR296661, KX090926, KT593874, KX061462).

2.2. Screening for ESBL production and carbapenem resistance

For phenotypic identification of ESBL producers, cefotaxime (30 µg) with and without clavulanic acid (30 µg/10 µg) and ceftazidime (30 µg) with and without clavulanic acid (30 µg/10 µg) were used in the combined disc method [15]. Once the zone of inhibition between the combination disc and the corresponding single antibiotic disc differed by ≥5 mm, the isolate was identified as an ESBL producer. The isolates were further confirmed as ESBL producers using the E-test and the MIC were determined for cefpodoxime, cefotaxime, ceftazidime, and ceftriaxone for all the phenotypically confirmed ESBL producers using the Hi-comb MIC test strip (HiMedia, Mumbai, India). The ESBL producing isolates were further screened for antibiotic susceptibility pattern against amikacin (AN, 30 µg), aztreonam (ATM, 30 µg), chloramphenicol (C, 30 µg), ceftriaxone (CRO, 30 µg), cefpodoxime (CPD, 10 µg), ceftazidime (CAZ, 30 µg), cefotaxime (CTX, 30 µg), cefepime (FEP, 30 µg), cefixime (CFM, 5 µg), cefoxitin (FOX, 30 µg), enrofloxacin (ENR, 5 µg), sulfamethoxazole/ trimethoprim (SXT, 23.75/1.25 µg), tetracycline (TE, 5 µg). The *E. coli* isolates were also tested for carbapenem resistance against imipenem (IPM, 10 µg), ertapenem (ETP, 10 µg) and meropenem (MRP, 10

Table 1
Details of the samples collected from the different farms of India.

Location	Longitude	Latitude	Farm holding	Number of piglets available at the time of sampling	Number of samples collected from piglets	Sex		Age (in month)			Health status		Breed		Number of samples collected from piglet farm workers
						Male	Female	< 1	1 to 2	2 to 3	Non-diarrhoeic	Diarrhoeic	Pure	Cross	
Bareilly, Uttar Pradesh	79.4320	28.3930	120-250	60	30	18	12	10	11	9	12	18	0	30	6
Aligarh, Uttar Pradesh	78.0888	27.8974	25-80	80	41	22	19	15	12	14	3	38	4	37	0
Chennai farm -1, Tamilnadu	80.2437	13.1625	60-125	60	28	12	16	10	18	0	13	15	16	12	10
Chennai farm -2, Tamilnadu	80.1041	13.0468	80-140	80	21	15	6	21	0	0	5	16	6	15	0
Hassan, Karnataka	76.1279	13.0210	50-160	60	35	21	14	15	20	0	10	25	21	14	5
Total					155	88	67	71	61	23	43	112	47	108	21

µg) along with EDTA disc (~1900 µg) for keyhole reaction to detect MBL isolates. Clinical and Laboratory Standards Institute (CLSI), 2014 breakpoints were used for interpretation of the susceptibility pattern [16]. The carbapenem-resistant isolates were subjected to the determination of minimum inhibitory concentration (MIC) by using a meropenem E-test strip (HiMedia, Mumbai, India).

2.3. Genotypic detection of antimicrobial resistance genes

The genomic DNA was extracted from ESBL positive isolates using QIAamp DNA Minikit (Qiagen, New Delhi, India) and PCR was carried out in a final reaction volume of 25 µL using 0.2 mL thin-wall PCR tubes (HiMedia, Mumbai, India). A master mix for a minimum of 10 samples was prepared and aliquoted in 23 µL (2x PCR mastermix-12.5 µL, DNase free water-8.5 µL, Forward primer-1.0 µL, Reverse primer-1.0 µL) quantities in each PCR tube. Two µL sample DNA was added to each tube to make the final volume of 25 µL. The primers and cycling conditions used for genotypic detection of selected antimicrobial resistance genes were given in supplementary Table 1.

2.4. DNA sequencing, plasmid replicon, and multilocus sequence (MLST) typing

The representative PCR products were purified using a gel purification kit (Qiagen, New Delhi, India) for sequencing. For antimicrobial resistance genes, the homology search was made using the BLAST algorithm (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). The plasmid DNA was extracted from blaNDM positive *E. coli* (n = 3) isolated from piglets and farmworkers using the Qiagen miniprep kit (Qiagen, Hilden, Germany). The plasmids were also tested by PCR for the presence of AMR genes. Plasmids were assigned to incompatibility groups by using the multiplex PCR [17]. The MLST against seven housekeeping genes for blaNDM positive *E. coli* isolates (n = 3) of piglets and humans were carried out by PCR [18] and sequenced by the Sanger method (Eurofins India Ltd, Bengaluru). The sequences were edited by using BioEdit v7.0.5 and queried in the MLST website (https://enterobase.warwick.ac.uk/species/ecoli/allele_st_search) for the allelic profile, sequence type and clonal complex.

2.5. Statistical analysis

The association of piglet's age, sex, breed with ESBL producing *E. coli* was tested by Chi-square Fisher's exact (two-tailed) test using SPSS version 20.0 statistical software (IBM Corp., Armonk, NY, USA). The Fisher's exact test was used when expected cell frequencies were <5. Antibiotic susceptibility phenotypic profiles and gene presence was converted into binary code. For each antibiotic, 0 denoted susceptibility and 1 denoted resistance similarly the presence of a gene was designated as 1 and absence as 0. Heatmap with hierarchical clustering of individual ESBL producing *E. coli* from piglets and farmworkers were performed using heatmaply package in R statistical program. Principal component analysis (PCA) was performed using the package Factoextra from the open-source statistical program R [19].

3. Results

From the questionnaire survey, it was observed that the farms in this study used beta-lactam and cephalosporin antibacterials for the treatment of individual sick animals. In all the farms, neither pen treatment nor antibiotics were mixed in feed. All the farms had cement flooring with well-ventilated animal shed. In all the farms, weaning was practiced between 35–42 days. No outbreak of contagious disease was recorded in the past 12 months. The common health complaints were piglet diarrhoea, wounds and abortion. The farm size ranged from 50–200 adult pigs. Faecal samples (n = 155) from both diarrhoeic (n = 112) and non diarrhoeic piglets (n = 43), <3 months of age along with

healthy farmworker's faecal samples (n = 21) were collected from five organized pig farms (Table 1). From piglet faecal samples, 124 *E. coli* (non-diarrhoeic, n = 102; diarrhoeic, n = 22) and from pig farmworkers, 21 *E. coli* were isolated on MacConkey agar supplemented with 1 mg/L cefotaxime. Of the 124 *E. coli* from piglets, 55 were identified as ESBL producers using the combined disk diffusion method. The antibiotic susceptibility testing of the ESBL *E. coli* (n = 55) isolated from piglets showed 100 % of the isolates tested were resistant to cefotaxime, cefpodoxime, ceftazidime, cefoxitin and higher extent to tetracycline (88.8 %, 8/9), amikacin (78.2 %, 44/55), sulfamethoxazole/ trimethoprim (78.2 %, 43/55), cefepime (76.4 %, 42/55), enrofloxacin (45.5 %, 25/55), chloramphenicol (77.7 %, 7/9), ertapenem (16.4 %, 9/55), imipenem (14.5 %, 8/55), meropenem (14.5 %, 8/55) and aztreonam (10.9 %, 6/55).

Of the ESBL positive isolates (n = 55), 24, 16 and 15 isolates were harbored by <1, 1–2 and 2–3 months of age piglets, respectively. A significantly higher number of ESBL producing *E. coli* isolates were recovered from < 1 month and 2–3 months of age piglets (p < 0.05). Crossbred piglets harbored 39 isolates while purebred carried 19 ESBL producing *E. coli*. The majority of ESBL producing *E. coli* isolates were isolated from diarrhoeic (n = 49) compared to non-diarrhoeic (n = 6) piglets. Diarrhoeic (50 %, 49/98) and cross-bred piglets (52.7 %, 39/74) harbored a significantly higher number of ESBL producing isolates (p < 0.05) than non-diarrhoeic (23 %, 6/26) and purebred (32 %, 16/50) piglets (Table 2). Carbapenem susceptibility screening of *E. coli* isolates of piglets (n = 9) showed resistance to imipenem (88.8 %, 8/9), ertapenem (66.6 %, 6/9) and meropenem (88.8 %, 8/9). Two isolates from piglets and one from farmworkers showed keyhole between EDTA and carbapenem discs indicating metallo beta-lactamase production. The majority of carbapenem resistant *E. coli* isolates were carried on diarrhoeic (n = 7) compared to non-diarrhoeic (n = 2) piglets. Of the two carbapenem resistant keyhole positive isolates each one harbored NDM-1 and NDM-5 variants.

Of 21 *E. coli* from pig farmworkers, nine and two isolates were ESBL producers and carbapenem resistant, respectively. The screening of ESBL *E. coli* (n = 9) isolated from farmworkers showed 100 % resistance to ceftazidime, cefotaxime, cefoxitin, cefpodoxime and higher extent to sulfamethoxazole/trimethoprim (88.8 %, 8/9), tetracycline (88.8 %, 8/9), chloramphenicol (77.7 %, 7/9), imipenem (22.2 %, 2/9), meropenem (22.2 %, 2/9), ertapenem (11.1 %, 1/9), moxalactam (22.2 %, 2/9) and aztreonam (11.1 %, 1/9).

On genotypic screening, the ESBL positive isolates from piglets (n = 55) harbored blaCTX-M (n = 11) and other antibiotic resistance genes like blaTEM-1 (n = 17), blaAmpC (n = 14), qnrS (n = 9), qnrA (n = 3), qnrB (n = 11), sul1 (n = 14), tetA (n = 12), tetB (n = 8). While, ESBL positive isolates from pig farmworkers (n = 9) harbored blaCTXM (n = 1) and other antibiotic resistance genes like, blaTEM (n = 5), blaAmpC (n = 6), qnrS (n = 4), qnrB (n = 2), sul1 (n = 2), tetA (n = 1) and tetB (n = 2). Two carbapenem resistant isolate from piglets (41/Aligarh/piglet, 25/CARI/piglet) and one isolate from pig farmworker (5/CARI/Human) carried blaNDM-5 (GenBank accession number: MF 044035) gene as well as blaCTXM and sul1 (Table 3).

In cluster, binary factors (such as antibiotics or genes) indicating presence as red (relative response 1) or absence as light grey (relative response 0). The heat map with cluster analysis showed most of the piglets and farm worker's isolates clustered together based on phenotypic resistance and AMR genes (Fig. 1). The ESBL producing *E. coli* of the farmworkers and piglets were divided into two groups in the principal component analysis (PCA) (Fig. 2). The PCA analysis suggested the phenotypic susceptibility pattern of ESBL positive isolates showed a positive association with the resistance profile. However, the AMR and virulence genes in ESBL positive isolates showed a negative association.

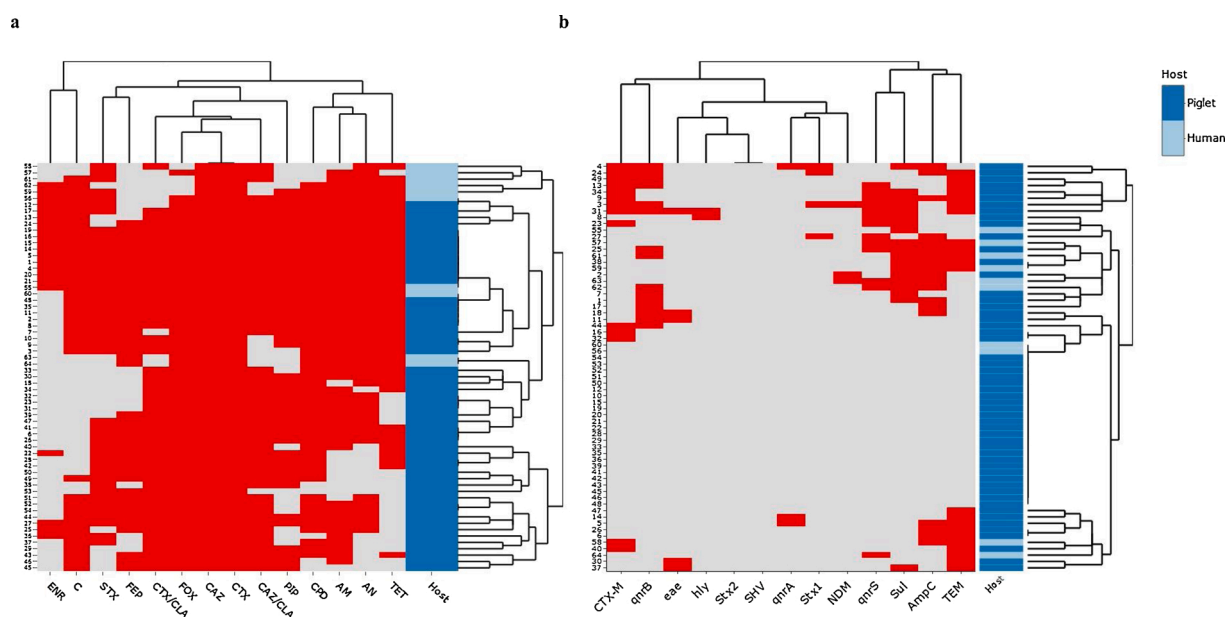
The plasmid replicon typing of the carbapenem resistant *E. coli* isolated from piglets (25/CARI/piglet) and pig farmworkers (5/CARI/Human) of the same farm revealed an IncX3 plasmid carrying blaNDM-5, blaCTXM, and sul1 genes. However, carbapenem resistant *E. coli*

Table 2Details of the ESBL producing *E. coli* isolates from piglets and pig farmworkers (number in the parenthesis indicates the number of *E. coli* isolates).

Location	Number of ESBL producing isolates from piglets	Sex		Age (in month)			Health status		Breed		Number of ESBL producing isolates from farm workers
		Male	Female	< 1	1 to 2	2 to 3	Non-diarrhoeic	Diarrhoeic	Pure	Cross	
Bareilly, Uttar Pradesh	9 (30)	5 (18)	4 (12)	1 (10)	2 (11)	6 (9)	2 (9)	7 (21)	0 (6)	9 (24)	8 (9)
Aligarh, Uttar Pradesh	26 (35)	14 (18)	12 (17)	11 (12)	6 (9)	9 (14)	2 (4)	26 (31)	0 (2)	26 (33)	0 (0)
Chennai farm -1, Tamilnadu	8 (21)	2 (9)	6 (12)	2 (6)	6 (15)	0 (0)	0 (9)	6 (12)	6 (14)	2 (7)	6 (7)
Chennai farm -2, Tamilnadu	7 (14)	3 (10)	4 (4)	7 (14)	0 (0)	0 (0)	0 (0)	7 (14)	6 (11)	1 (3)	0 (0)
Hassan, Karnataka	5 (24)	4 (14)	1 (10)	3 (10)	2 (14)	0 (0)	2 (4)	3 (20)	4 (17)	1 (7)	5 (5)
Total	55 (124)	28 (69)	27 (55)	24 (52)	16 (49)	15 (23)	6 (26)	49 (98)	16 (50)	39 (74)	19 (21)
P value		0.36		<0.05			0.02*		0.03*		

Table 3Antibiotic resistance determinants and plasmid type of NDM positive *E. coli* isolates.

S. No	Isolate	Location	Host	Sex	Age	Health status	Antibiotic resistance determinants	Plasmid type
1	5/CARI/Human	Bareilly, Uttar Pradesh	Human	Male	28 years	healthy	<i>bla</i> NDM-5, <i>bla</i> CTX-M, <i>su</i> 1	IncX3
2	25/CARI/piglet	Bareilly, Uttar Pradesh	Piglet	Female	31-60 days	diarrhoeic	<i>bla</i> NDM-5, <i>bla</i> CTX-M, <i>su</i> 1, <i>qnr</i> S	IncX3
3	41/Aligarh/piglet	Aligarh, Uttar Pradesh	Piglet	Female	0-30 days	diarrhoeic	<i>bla</i> NDM-1, <i>bla</i> TEM-1, <i>bla</i> CTX-M, <i>su</i> 1, <i>qnr</i> B	Inc11

**Fig. 1.** Heatmap of individual ESBL producing *E. coli* from piglets and farmworkers showing hierarchical clustering of isolates and factors. Binary factors (such as antibiotics or genes) indicating presence as red (relative response 1) or absence as light grey (relative response 0). The left side panel indicates the antibiotic susceptibility pattern and the right side panel indicates the antimicrobial resistance genes.

isolate from piglet (41/Aligarh/piglet) of Aligarh farm revealed an Inc11 plasmid carrying the *bla*NDM-1 (Table 3). The multilocus sequence analysis of the NDM positive isolates from piglets and farmworkers exhibited the sequence type (ST)-167.

4. Discussion

This study identified piglets and pig farmworkers as a potential source of extended spectrum beta-lactam and carbapenem resistant *E. coli*. The higher occurrence of ESBL producing *E. coli* isolates in piglets (44.4%, 55/124) and farmworkers (90.5%, 19/21) might be associated either with the selection of *E. coli* isolates against cefotaxime on

MacConkey agar or with the use of beta-lactam and cephalosporin antibiotics on the farms investigated. Similarly, Mollenkopf et al. screened pig farm environmental and pig samples on ESBL selective medium and observed higher ESBL positives [20]. The screening of pooled faeces and environmental samples from 58 pig farms in Germany with MacConkey agar with cefotaxime 1 µg/mL yielded ESBL and carbapenem resistant *E. coli* isolates with *bla*VIM-1 gene [21]. In the present study, the ESBL producers were 100 % resistant to cefotaxime, ceftazidime, ceftazidime. A similar resistance pattern was noticed in ESBL positive isolates [9]. Interestingly, nine isolates from piglets showed resistance to one or more carbapenem antibiotics. In our earlier study, the significance of age was noted among carbapenem resistant *E. coli* isolated from

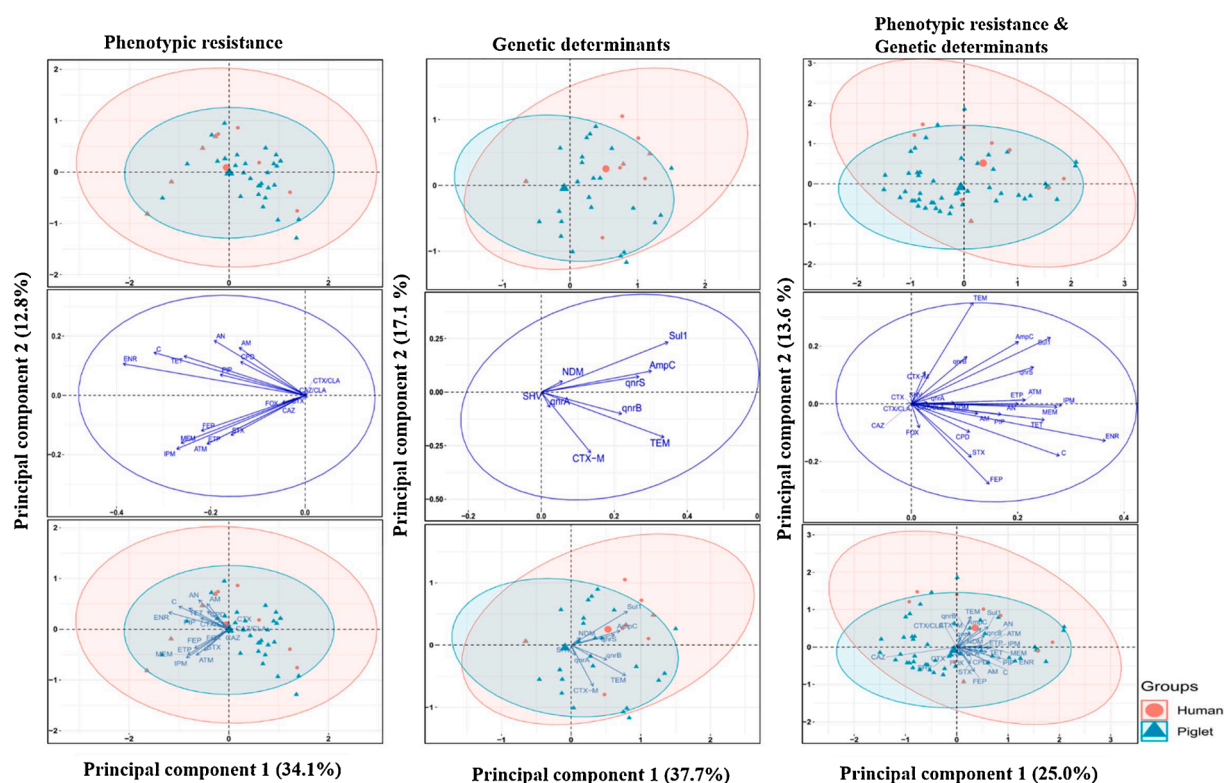


Fig. 2. Principal Component Analysis (PCA) plots of ESBL producing *E. coli* isolates. The ESBL *E. coli* are encompassed in 95% confidence intervals grouping based on source of the isolate (human and piglets).

piglets [8]. In calves also, carbapenem resistance was noted in the younger age group and diarrhoeal calves harbored higher rates of meropenem ($p < 0.05$) and imipenem ($p < 0.001$) resistant isolates than healthy calves [22]. In the present study, significantly, a higher number of ESBL producing *E. coli* were recovered from < 1 month and 2-3 months of age piglets. Of nine carbapenem resistant *E. coli* from piglets, two harbored *bla*NDM-1. In India, carbapenem resistant *E. coli* harboring NDM-1 and OXA-48 enzymes were reported in younger age group piglets [8–9]. Kong et al. screened anal swabs of pigs from China and reported *bla*NDM-5 carbapenemase genes [23]. A systematic review reported the carbapenem-resistant Enterobacterales in wildlife, food-producing, and companion animals [24].

In this study, the ESBL isolates were also resistant to other classes of antibiotics. The probable reason might be that extended spectrum beta-lactamases are associated with co-resistance to other antibacterial classes [25]. In addition, the present study reported the ESBL positive isolates carrying other resistance genes. Similarly, Liu et al. reported ESBL producing *E. coli* isolates co-harboring β -lactamase and plasmid-mediated quinolone resistance (PMQR) genes [26]. Studies from India indicated, co-harboring of PMQR and other resistance genes in ESBL producing and carbapenem resistant *E. coli* isolated from piglets and dog [8–10]. The PCA and hierarchical cluster analysis showed an overlapping cluster of the resistant isolates of farmworkers and piglets indicated co-sharing of resistance. The carbapenem resistant isolates from piglets and farmworkers of the same farm shared common resistant genes (*bla*NDM-5, *bla*CTX-M, *su1*), sequence (ST-167) and plasmid type (IncX3) indicated a potential common source of infection or cross-species transmission.

The interview with the *bla*NDM-5 positive farmworker revealed no past medical history of major diseases and hospitalization, however, acknowledged frequent visits to clinician for minor ailments such as cold and fever. Fischer et al. demonstrated 6 % carriage rates of ESBL producing *E. coli* among pig farmers and molecular typing results showed the possible exchange of resistance determinants between human and

environmental isolates [27]. Carbapenemase producing genes are primarily plasmid-mediated and co-resistance is an important issue [28]. Recently in calves, *E. coli* positive for *bla*VIM-1 carbapenemase enzyme and Inc11 plasmid was reported from India [22]. Carbapenemase producing *E. coli* with *bla*IMP-27 on an IncQ1 plasmid was reported from the environment of swine farms in USA [20]. *Escherichia coli* ST-167 with IncX3 plasmid carrying *bla*NDM-7 in addition to *bla*TEM-1 and *bla*CMY-42 genes were reported from clinical blood specimens of gallbladder carcinoma patient of India [29]. Dahms et al. reported the occurrence of ESBL producing *E. coli* in cattle faecal sample and farmworkers with identical MLST sequence type (ST) 3891 and CTX-M allele [30]. The dominant ESBL genes in animals were *bla*CTX-M-1, while in humans the common variants were *bla*CTX-M-15 and *bla*CTX-M-14. However, recent studies showed that the prevalence of CTX-M-1 genes in humans [31] and rare identification of CTX-M-15 in animal isolates [32]. Dohmen et al. reported that the human and pig isolates of the two farms carried comparable ESBL gene types and similar sequence and plasmid types [33].

The present study demonstrated both the animal and human *E. coli* isolates shared common AMR genes, ST and plasmid type signify the possible cross-species transfer or common environmental source. Reducing the dissemination and transmission of resistant bacteria within and between animals, human populations and the environment is crucial to combat AMR.

5. Conclusion

Despite the small sample size, the study highlighted faecal carriage and co-sharing of extended-spectrum beta-lactamase (ESBL) and New Delhi Metallo beta-lactamase (NDM) producing *E. coli* between piglets and pig farmworkers. Hence, epidemiological surveillance of ESBL producing and carbapenem resistance in pigs is required to limit the transfer of resistance.

Declaration of Competing Interest

All authors declare that there are no financial or other relationships that might lead to a conflict of interest. All authors have seen and approved the manuscript and have contributed significantly to the work.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.cimid.2020.101564>.

References

- [1] P. Nordmann, M. Gniadkowski, C.G. Giske, et al., Identification and screening of carbapenemase-producing Enterobacteriaceae, *Clin Microbiol Infect.* 18 (2012) 432–438.
- [2] World Health Organization, Antimicrobial resistance: global report on surveillance, World Health Organization, Geneva, Switzerland, 2014.
- [3] P. Nordmann, L. Poirel, T.R. Walsh, et al., The emerging NDM carbapenemases, *Trends Microbiol.* 19 (12) (2011) 588–595.
- [4] A.J. Stewardson, A. Allignol, J. Beyersmann, et al., The health and economic burden of bloodstream infections caused by antimicrobial-susceptible and non-susceptible Enterobacteriales and *Staphylococcus aureus* in European hospitals, 2010 and 2011: a multicentre retrospective cohort study, *Euro. surveill.* (21) (2016) 1–12.
- [5] H. Gelband, P. Molly Miller, S. Pant, et al., The state of the world's antibiotics 2015, *Wound Healing Southern Africa.* (8) (2015) 30–34.
- [6] J. Fischer, I. Rodríguez, S. Schmöger, et al., *E. coli* producing VIM-1 carbapenemase isolated on a pig farm, *J Antimicrob Chemother.* 67 (7) (2012) 1793–1795.
- [7] S. Ghatak, A. Singha, A. Sen, et al., Detection of New Delhi Metallo-beta-Lactamase and Extended-Spectrum beta-Lactamase Genes in *Escherichia coli* Isolated from Mastitic Milk Samples, *Transbound emerg dis.* 60 (5) (2013) 385–389.
- [8] B.S. Pruthivishree, O.R. Vinodh Kumar, D.K. Sinha, et al., Spatial molecular epidemiology of carbapenem-resistant and New Delhi metallo beta-lactamase (*bla*NDM) producing *Escherichia coli* in the piglets of organized farms in India, *J appl microbiol.* 122 (2017) 1537–1546.
- [9] K.R. Nirupama, O.R. Vinodh Kumar, B.S. Pruthivishree, et al., Molecular characterization of *bla*OXA-48 carbapenemase-, extended-spectrum β -lactamase- and Shiga toxin-producing *Escherichia coli* isolated from farm piglets in India, *J Glob Antimicrob resist.* 13 (2018) 201–205.
- [10] B.S. Pruthivishree, O.R. Vinodh Kumar, M. Sivakumar, et al., Molecular characterization of extensively drug resistant (XDR), extended spectrum beta-lactamases (ESBL) and New Delhi Metallo beta-lactamase-1 (*bla*NDM1) producing *Escherichia coli* isolated from a male dog-a case report, *Veterinarski arhiv.* 88 (2018) 139–148.
- [11] World Health Organization, Global priority list of antibiotic-resistant bacteria to guide research, discovery, and development of new antibiotics, Available at: 2017 <http://www.who.int/medicines/publications/global-priority-list-antibiotic-resistant-bacteria/en/>.
- [12] R.S. Singer, R. Finch, H.C. Wegener, et al., Antibiotic resistance – the interplay between antibiotic use in animals and human beings, *Lancet Infect Dis.* 3 (2003) 47–51.
- [13] O.R. Vinodh Kumar, B.R. Singh, D.K. Sinha, et al., Risk factor analysis, antimicrobial resistance and pathotyping of *Escherichia coli* associated with pre- and post-weaning piglet diarrhoea in organised farms, India, *Epidemiol Infect.* (2019) 147.
- [14] O.R. Vinodh Kumar, S. Sircar, B.S. Pruthivishree, et al., Cross-sectional study on rotavirus A (RVA) infection and assessment of risk factors in pre- and post-weaning piglets in India, *Tropical Animal Health and Production.* (2019), <https://doi.org/10.1007/s11250-019-01999-8>.
- [15] J. Andrews, Detection of extended-spectrum β -lactamases (ESBLs) in *E. coli* and *Klebsiella* species, *J Antimicrob Chemother* (2003). Available at <http://bsac.org.uk/wp-content/uploads/2012/02/Ecoliklebsiella.pdf> (Accessed June 2018).
- [16] Clinical and Laboratory Standards Institute (CLSI), Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Fourth Informational Supplement 2014. M100-S24. Wayne, PA, USA: CLSI, Vol 34, 2014, pp. 56–60.
- [17] T.J. Johnson, Y.M. Wannemuehler, S.J. Johnson, et al., Plasmid replicon typing of commensal and pathogenic *Escherichia coli* isolates, *Appl Environ microbiol* 73 (2007) 976–1983.
- [18] S.Y. Tartof, O.D. Solberg, A.R. Manges, et al., Analysis of auropathogenic *Escherichia coli* clonal group by multilocus sequence typing, *J clin microbiol.* 43 (2005) 5860–5864.
- [19] R Core Team, R: A language and environment for statistical computing, URL, R Foundation for Statistical Computing, Vienna, Austria, 2014, <http://www.R-project.org/>.
- [20] D.F. Mollenkopf, J.W. Stull, D.A. Mathys, et al., Carbapenemase-producing Enterobacteriales recovered from the environment of a swine farrow-to-finish operation in the United States, *Antimicrob agents chemother* (2016) pp. AAC-01298.
- [21] N. Roschanski, A. Friese, C. von Salviati-Claudius, et al., Prevalence of carbapenemase producing Enterobacteriales isolated from German pig-fattening farms during the years 2011–2013, *Vet Microbiol.* 200 (2017) 124–129.
- [22] S.M. Murugan, D.K. Sinha, O.R. Vinodh Kumar, et al., Epidemiology of carbapenem-resistant *Escherichia coli* and first report of *bla*VIM carbapenemases gene in calves from India, *Epidemiol infect.* 147 (2019) e159.
- [23] L.H. Kong, C.W. Lei, S.Z. Ma, et al., Various sequence types of *Escherichia coli* isolates coharboring *bla*NDM-5 and *mcr-1* genes from a commercial swine farm in China, *Antimicrob Agents chemother.* 61 (3) (2017) e02167–16.
- [24] R. Kock, I. Daniels-Haardt, K. Becker, et al., Carbapenem-resistant Enterobacteriales in wildlife, food-producing, and companion animals: a systematic review, *Clin Microbiol Infect.* (2018).
- [25] D. Okoche, B.B. Asiimwe, F.A. Katabazi, et al., Prevalence and characterization of carbapenem-resistant Enterobacteriales isolated from Mulago National Referral Hospital, Uganda, *PloS one.* 10 (2015) p.e0135745.
- [26] X. Liu, H. Liu, L. Wang, et al., Molecular characterization of extended-spectrum β -lactamase-producing multidrug resistant *Escherichia coli* from swine in Northwest China, *Front microbiol* 9 (2018), <https://doi.org/10.3389/fmicb.2018.01756>.
- [27] J. Fischer, K. Hille, I. Ruddat, et al., Simultaneous occurrence of MRSA and ESBL-producing Enterobacteriales on pig farms and in nasal and stool samples from farmers, *Vet Microbiol.* 200 (2017) 107–113.
- [28] EFSA Panel on Biological Hazards (BIOHAZ), Scientific Opinion on Carbapenem resistance in food animal ecosystems, *EFSA Journal.* 11 (2013) 3501.
- [29] N.D. Ragupathi, D.M. Sethuvel, R. Gajendiran, et al., First Indian report of IncX3 plasmid carrying *bla*NDM-7 in *Escherichia coli* from bloodstream infection: potential for rapid dissemination, *New Microbes New infect.* 17 (2017) 65–68.
- [30] C. Dahms, N.O. Hübner, A. Kossow, et al., Occurrence of ESBL-producing *Escherichia coli* in livestock and farmworkers in Mecklenburg-Western Pomerania, Germany, *PLoS One.* 10 (2015) p.e0143326.
- [31] J.Y. Madec, M. Haenni, V. Métayer, et al., High prevalence of the animal-associated *bla*CTX-M-1 Inc11/ST3 plasmid in human *Escherichia coli* isolates, *Antimicrob Agents chemother.* 59 (9) (2015) 5860–5861.
- [32] S. Dahmen, M. Haenni, P. Chatre, et al., Characterization of *bla*CTX-M IncFII plasmids and clones of *E. coli* from pets in France, *J Antimicrob Chemother.* 68 (12) (2013) 2797–2801.
- [33] W. Dohmen, M.J. Bonten, M.E. Bos, S. van Marm, J. Scharringa, J.A. Wagenaar, D. J. Heederik, Carriage of extended-spectrum β -lactamases in pig farmers is associated with occurrence in pigs, *Clinical Microbiology and Infection.* 10 (2015) 917–923.

Further reading

- [34] L. Poirel, T.R. Walsh, V. Cuvillier, et al., Multiplex PCR for detection of acquired carbapenemase genes, *Diagn Microbial Infect dis* 70 (2020) 119–123.
- [35] M.B. Kerrn, T. Klemmensen, N. Primodt-Møller, et al., Susceptibility of Danish *E. coli* strains isolated from urinary tract infections and bacteraemia, and distribution of sul genes conferring sulphonamide resistance, *J Antimicrob Chemother.* 50 (2002) 513–516.
- [36] H. Ciesielczuk, M. Hornsey, V. Choi, et al., Development and evaluation of a multiplex PCR for eight plasmid-mediated quinolone-resistance determinants, *J Med Microbiol.* 62 (2013) 1823–1827.
- [37] C. Maynard, S. Bekal, F. Sanschagrin, et al., Heterogeneity among virulence and antimicrobial resistance gene profiles of extraintestinal *Escherichia coli* isolates of animal and human origin, *J Clin Microb.* 42 (2004) 5444–5452.